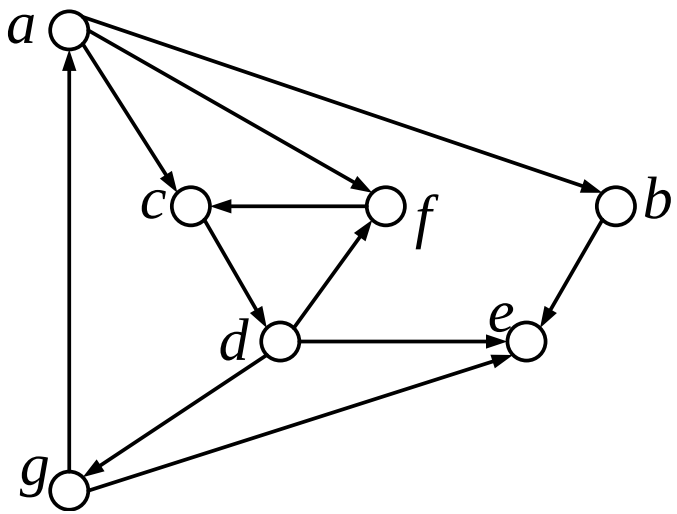
DFS and the Proof of White Path Theorem

Yufei Tao's Teaching Team

Department of Computer Science and Engineering
Chinese University of Hong Kong

Let's first go over the DFS algorithm through an example.

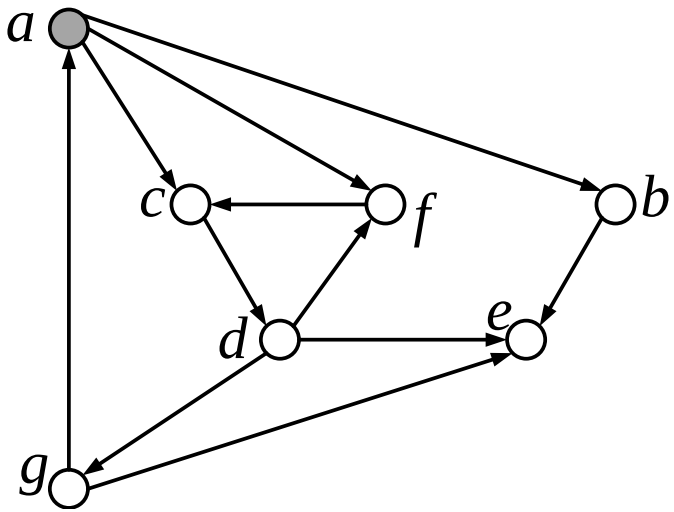
Input



Suppose we start from the vertex *a*, namely *a* is the root of DFS tree.

DFS

Firstly, set all the vertices to be white. Then, create a stack S, push the starting vertex a into S and color it gray. Create a DFS Tree with a as the root. We also maintain the time interval $I(u)$ of each vertex u.

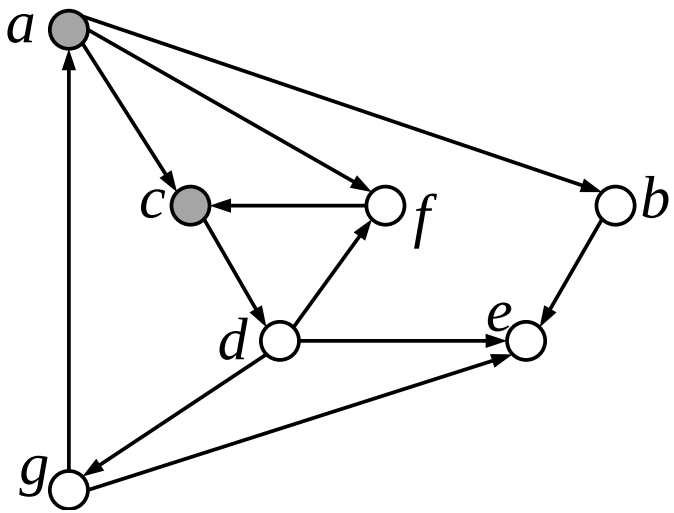


DFS Tree	Time Interval
a	$I(a) = [1, \ ]$

$S = (a)$.

DFS

Top of stack: a, which has white out-neighbors b, c, f. Suppose we access c first. Push c into S.

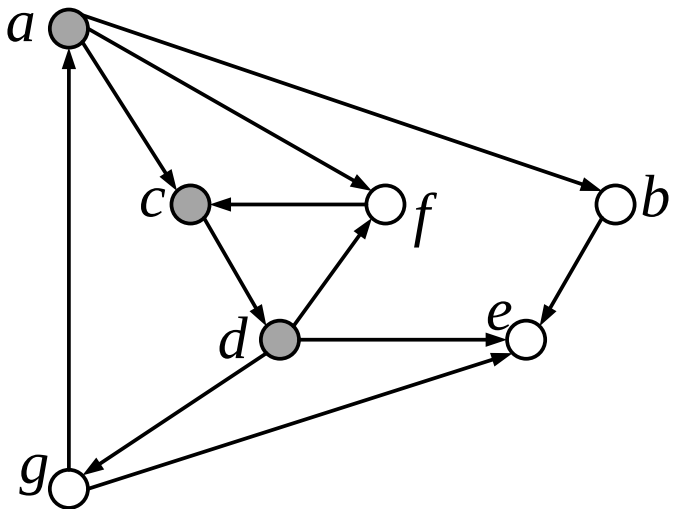


DFS Tree	Time Interval
a	$I(a) = [1, \ ]$
|	
c	$I(c) = [2, \ ]$

$S = (a, c)$.

DFS

After pushing d into S:

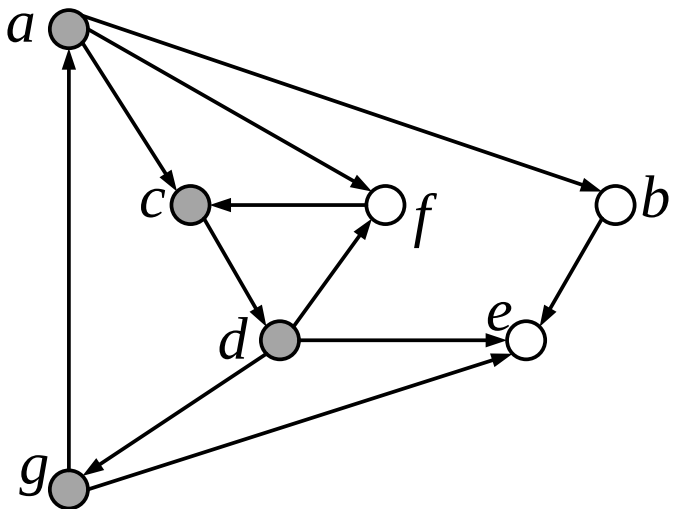


DFS Tree	Time Interval
a	$I(a) = [1, \ ]$
|	
c	$I(c) = [2, \ ]$
|	
d	$I(d) = [3, \ ]$

$S = (a, c, d)$.

DFS

Now d tops the stack. It has white out-neighbors e, f and g. Suppose we visit g first. Push g into S.



DFS Tree

a
|
c
|
d
|
g

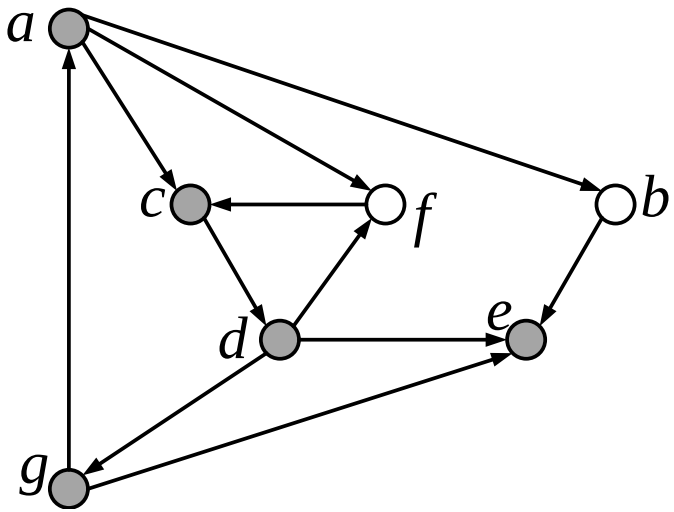
Time Interval

$I(a) = [1, \ ]$
$I(c) = [2, \ ]$
$I(d) = [3, \ ]$
$I(g) = [4, \ ]$

$S = (a, c, d, g)$.

DFS

After pushing e into S:

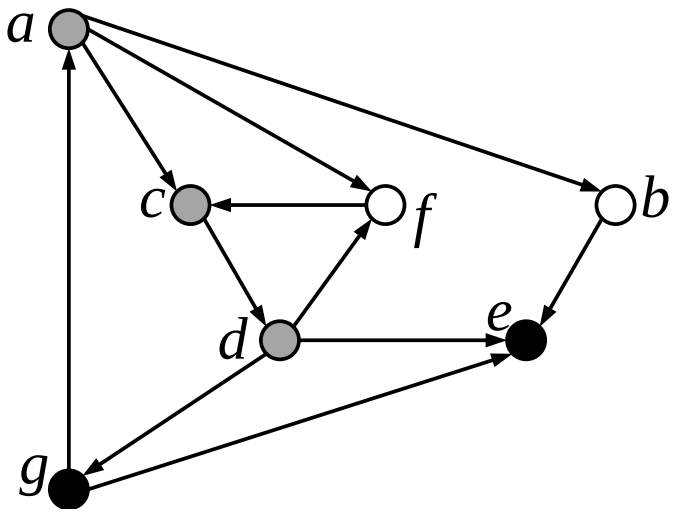


DFS Tree	Time Interval
a	$I(a) = [1, \ ]$
$\vert$	
c	$I(c) = [2, \ ]$
$\vert$	
d	$I(d) = [3, \ ]$
$\vert$	
g	$I(g) = [4, \ ]$
$\vert$	
e	$I(e) = [5, \ ]$

$S = (a, c, d, g, e)$.

DFS

e has no white out-neighbors. So pop it from S, and color it black.
Similarly, g has no white out-neighbors. Pop it from S, and color it black.

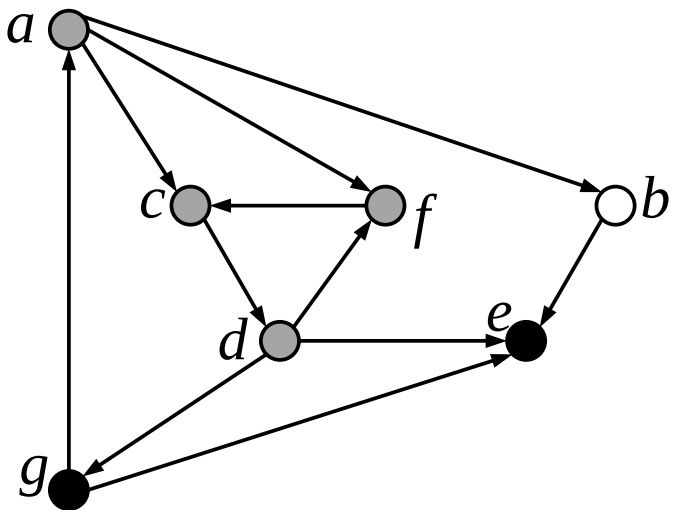


DFS Tree	Time Interval
a	$I(a) = [1, \ ]$
$\vert$	
c	$I(c) = [2, \ ]$
$\vert$	
d	$I(d) = [3, \ ]$
$\vert$	
g	$I(g) = [4, 7]$
$\vert$	
e	$I(e) = [5, 6]$

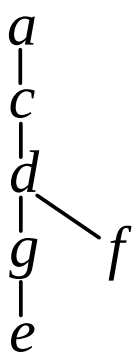
$S = (a, c, d)$.

DFS

Now d tops the stack again. It still has a white out-neighbor f. So, push f into S.



DFS Tree



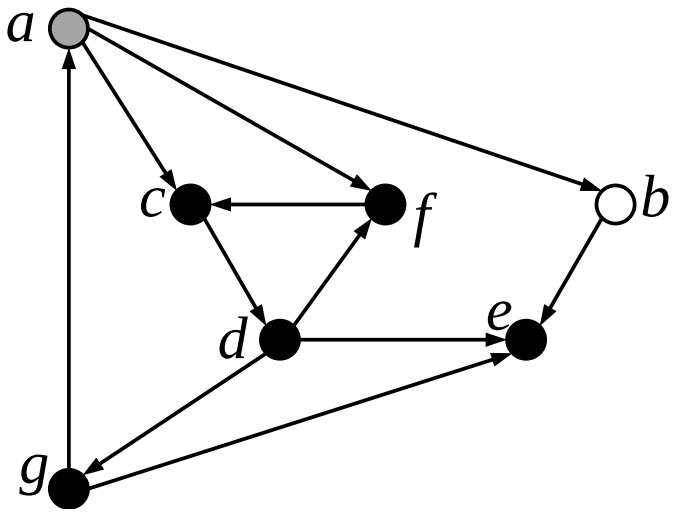
Time Interval

$I(a) = [1, \ ]$
$I(c) = [2, \ ]$
$I(d) = [3, \ ]$
$I(g) = [4, 7]$
$I(e) = [5, 6]$
$I(f) = [8, \ ]$

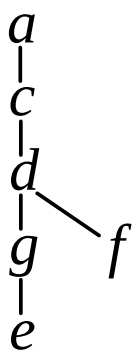
$S = (a, c, d, f)$.

DFS

After popping f, d, c:



DFS Tree



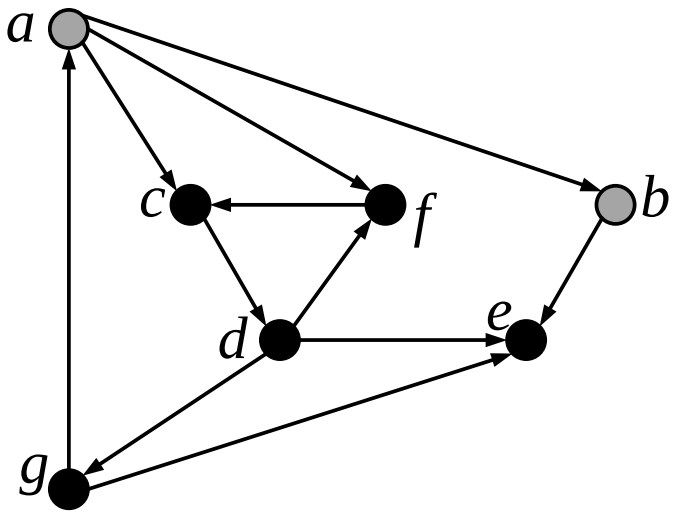
Time Interval

$I(a) = [1, \ ]$
$I(c) = [2, 11]$
$I(d) = [3, 10]$
$I(g) = [4, 7]$
$I(e) = [5, 6]$
$I(f) = [8, 9]$

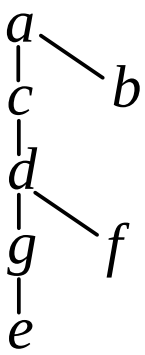
$S = (a)$.

DFS

Now a tops the stack again. It still has a white out-neighbor b. So, push b into S.



DFS Tree



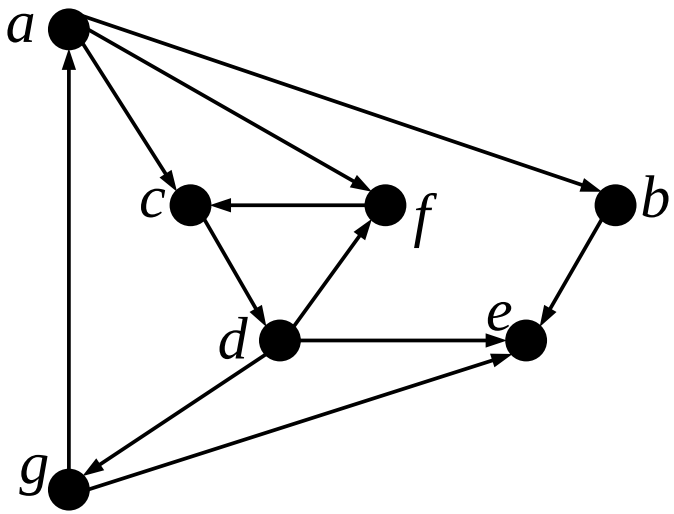
Time Interval

$I(a) = [1, \ ]$
$I(c) = [2, 11]$
$I(d) = [3, 10]$
$I(g) = [4, 7]$
$I(e) = [5, 6]$
$I(f) = [8, 9]$
$I(b) = [12, \ ]$

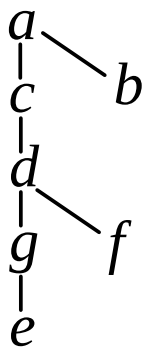
$S = (a, b)$.

DFS

After popping b and a:



DFS Tree



Time Interval

$I(a) = [1, 14]$
$I(c) = [2, 11]$
$I(d) = [3, 10]$
$I(g) = [4, 7]$
$I(e) = [5, 6]$
$I(f) = [8, 9]$
$I(b) = [12, 13]$

$S = ()$.

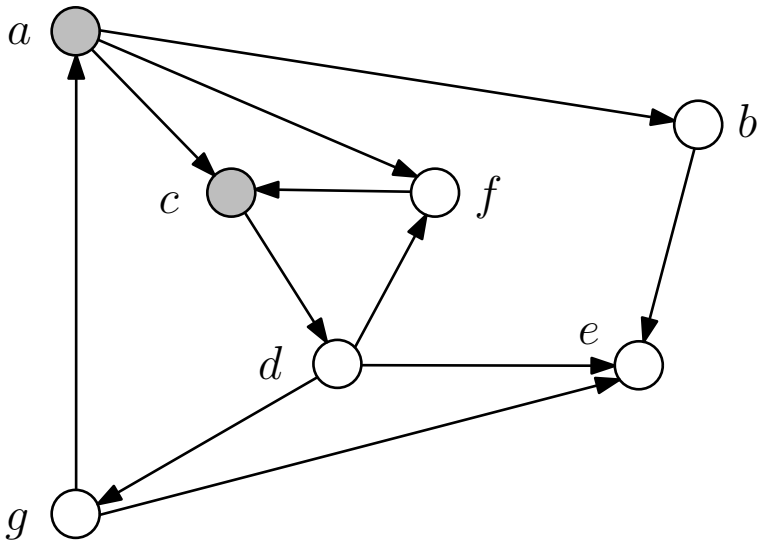
Now, there is no white vertex remaining, our algorithm terminates.

Recall:

White Path Theorem: Let u be a vertex in G. Consider the moment when u is pushed into the stack in the DFS algorithm. Then, a vertex v becomes a proper descendant of u in the DFS-forest **if and only if** the following is true:

we can go from u to v by travelling only on white vertices.

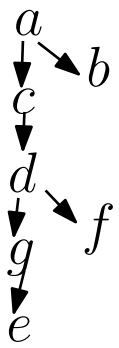
Example



DFS Tree



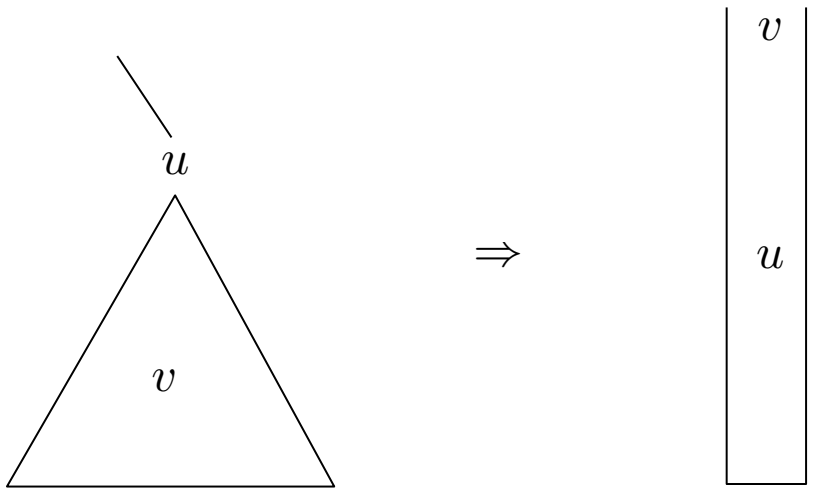
Final DFS Tree



$S =$ $\boxed{a \;\; c}$

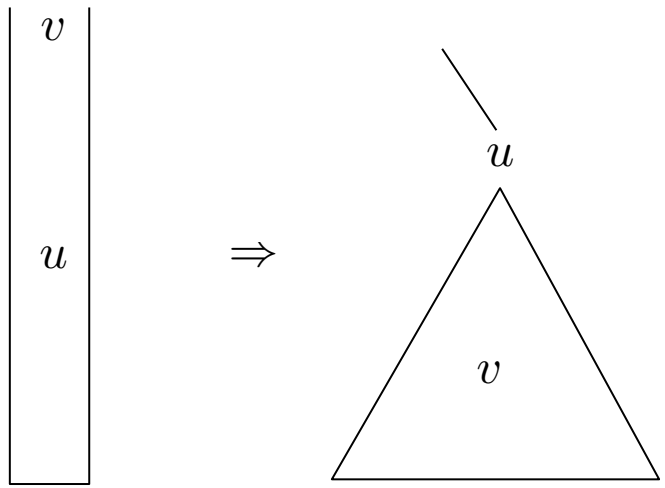
Lemma 1: Consider any two distinct vertices u and v in a DFS-tree. If v is a descendant of u in a DFS-tree, then v enters the stack while u is in the stack.

The proof is left to you.



Lemma 2: Consider any two distinct vertices u and v in a DFS-tree. If v enters the stack while u is in the stack, then v is a descendant of u in a DFS-tree.

The proof is left to you.



Proof of White Path Theorem

White Path Theorem: Let u be a vertex in G. Consider the moment when u is pushed into the stack in the DFS algorithm. Then, a vertex v becomes a proper descendant of u in the DFS-forest **if and only if** the following is true:

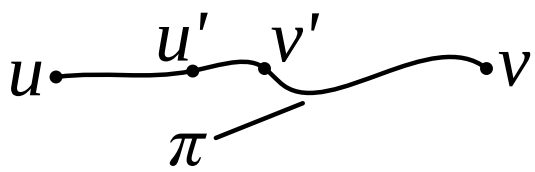
we can go from u to v by travelling only on white vertices.

Proof: The "only-if direction" ($\Rightarrow$): Let v be a descendant of u in the DFS tree. Let π be the path from u to v in the tree. By Lemma 1, all the nodes on π entered the stack after u. Hence, π must be white at the moment when u enters the stack.

Proof of White Path Theorem

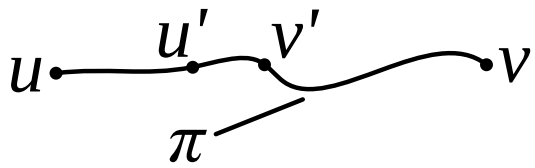
The "if direction" ($\Leftarrow$): When u enters the stack, there is a white path π from u to v. We will prove that all the vertices on π must be descendants of u in the DFS-forest.

Suppose that this is not true. Let v' be the first vertex on π — in the order from u to v — that is not a descendant of u in the DFS-forest. Clearly $v' \neq u$. Let u' be the vertex that precedes v' on π; note that u' is a descendant of u in the DFS-forest.



By Lemma 2, u' entered the stack after u.

Proof of White Path Theorem



Consider the moment when u' turns black (i.e., u' leaving the stack). Node u must remain in the stack currently (first in last out).

1. The color of v' cannot be white.

   Otherwise, v' is a white out-neighbor of u, which contradicts the fact that u' is turning black.

2. Hence, the color of v' must be gray or black.

   Recall that when u entered stack, v' was white. Therefore, v' must have been pushed into the stack while u was still in the stack. By the lemma on Slide 16, v' must be a descendant of u. This, however, contradicts the definition of v'.